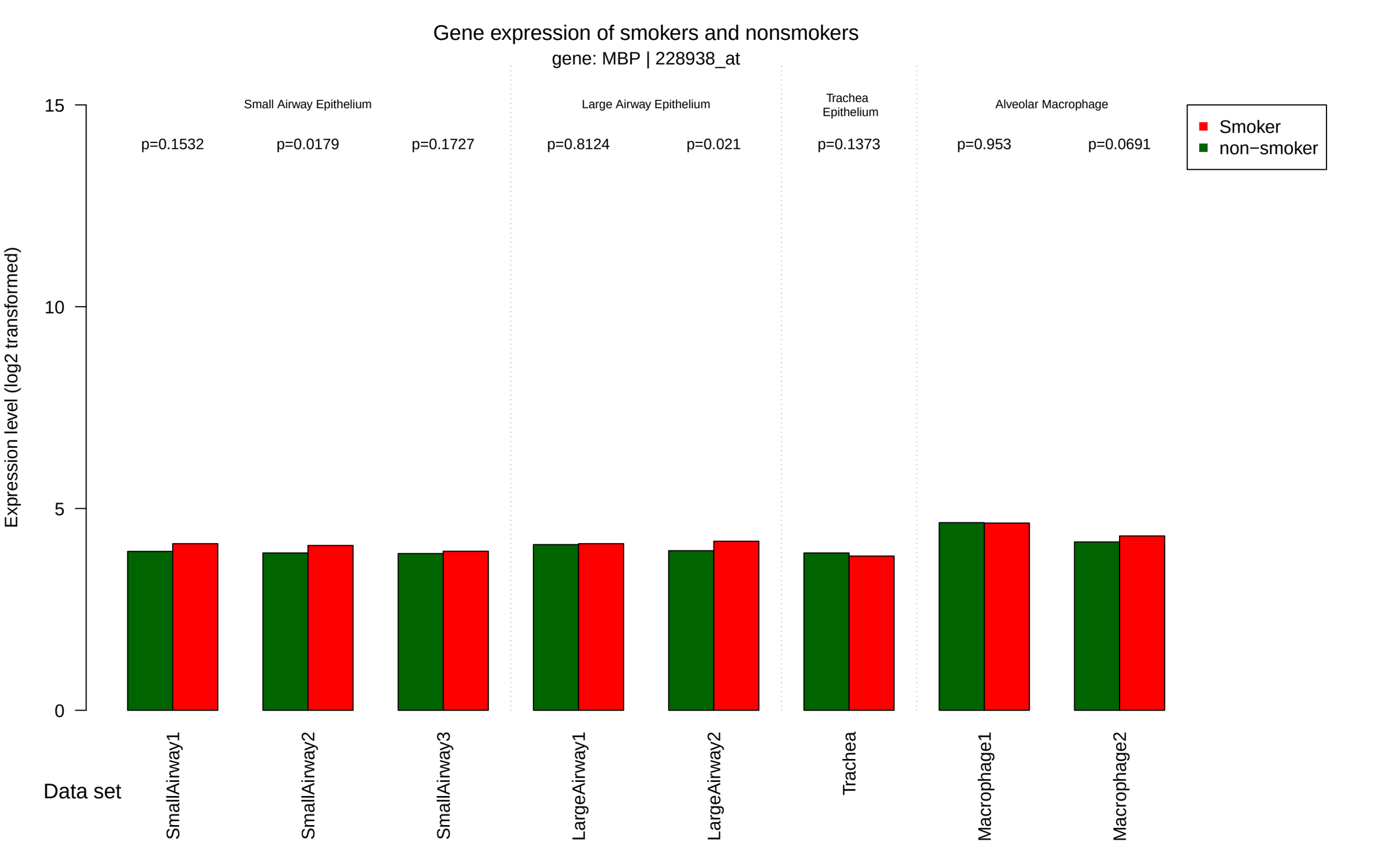
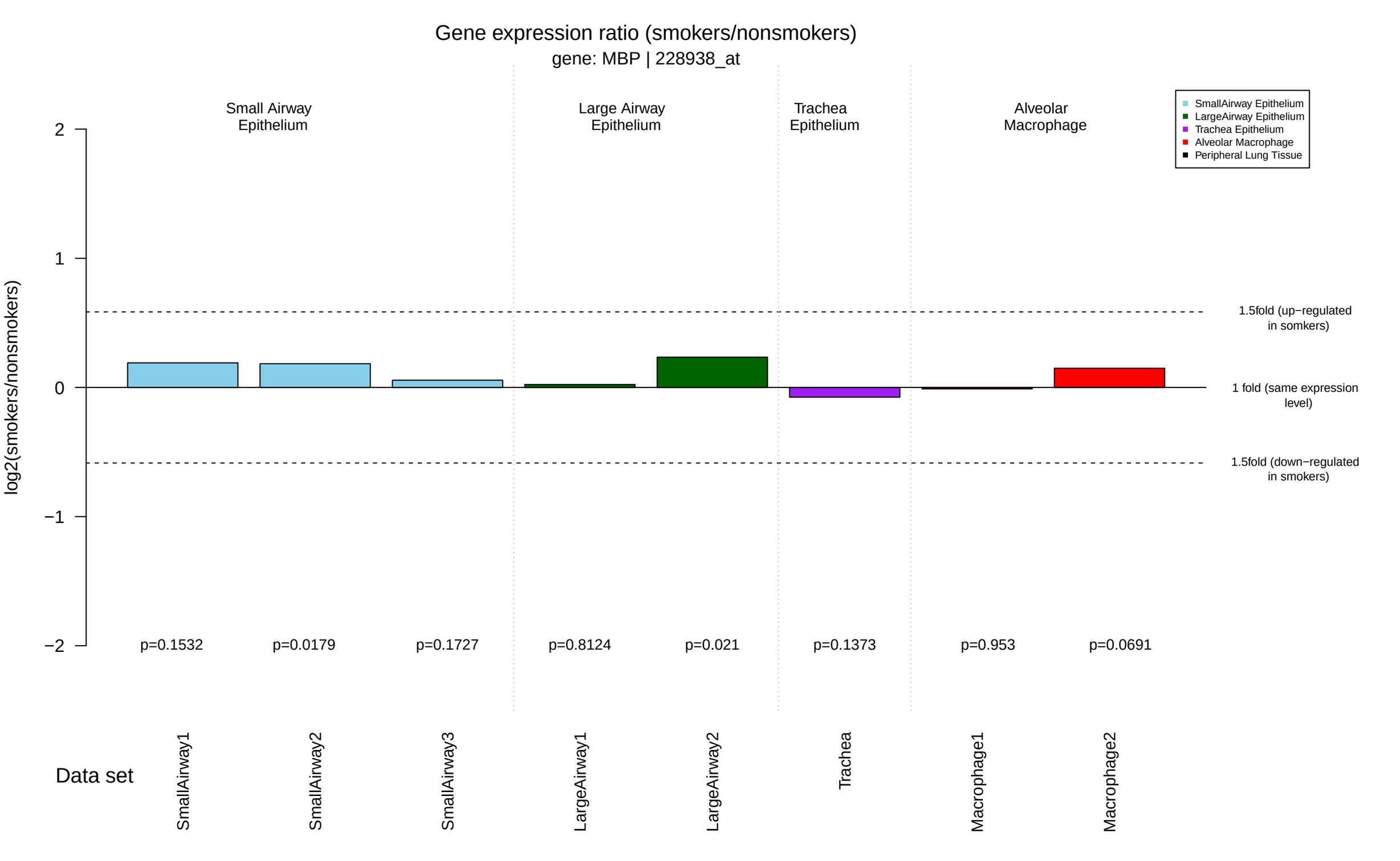
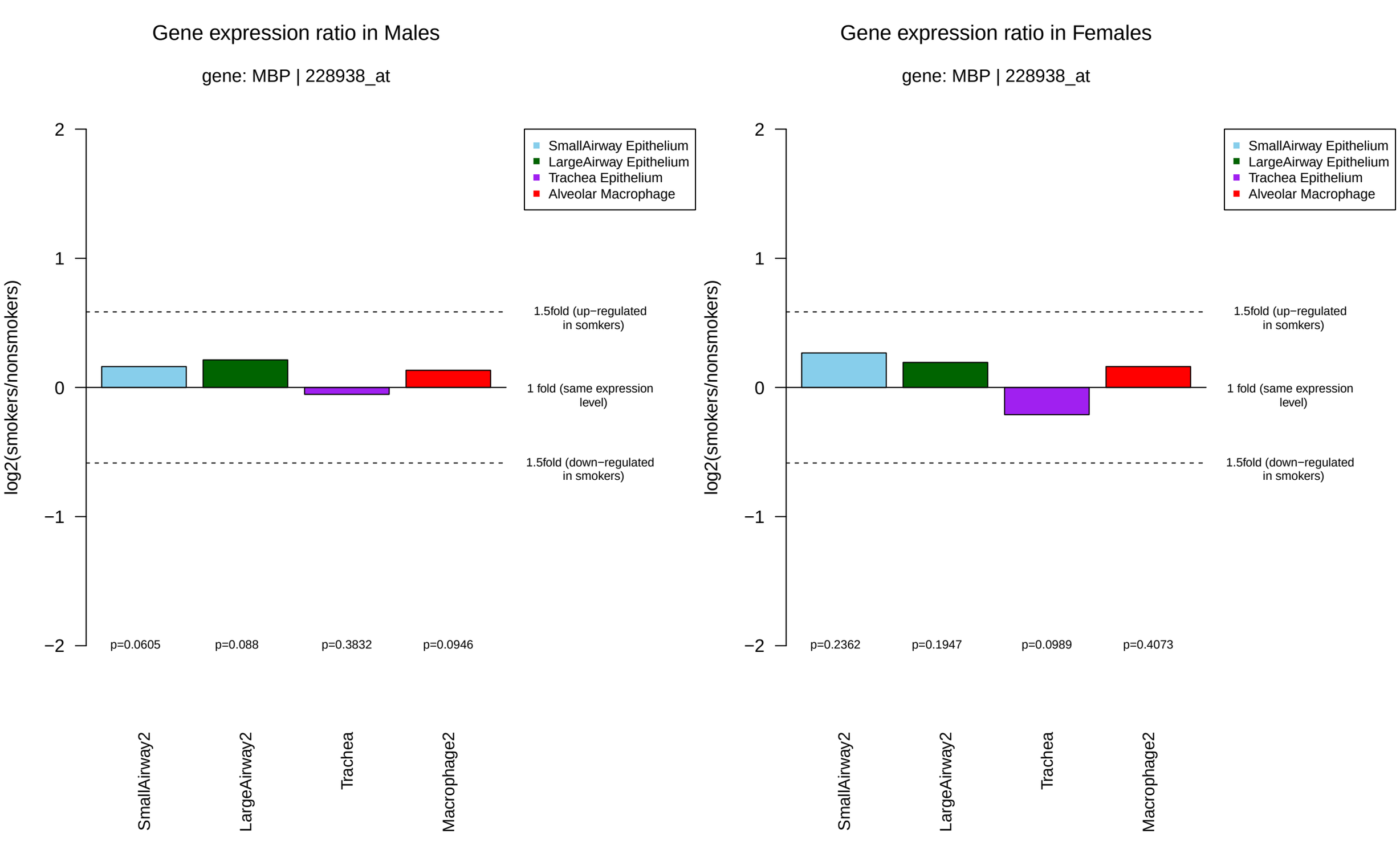
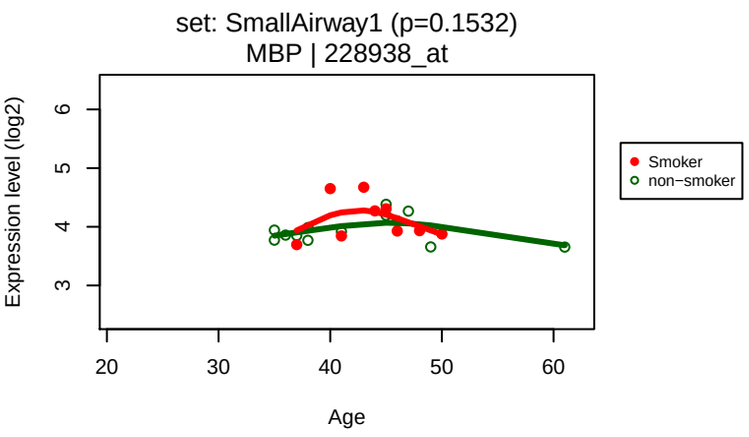
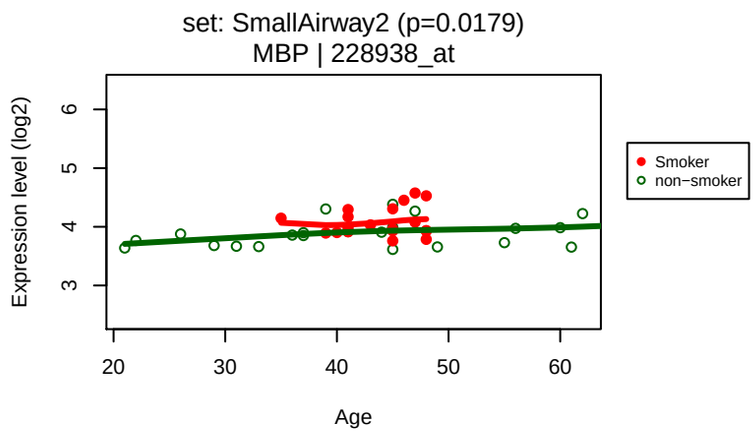
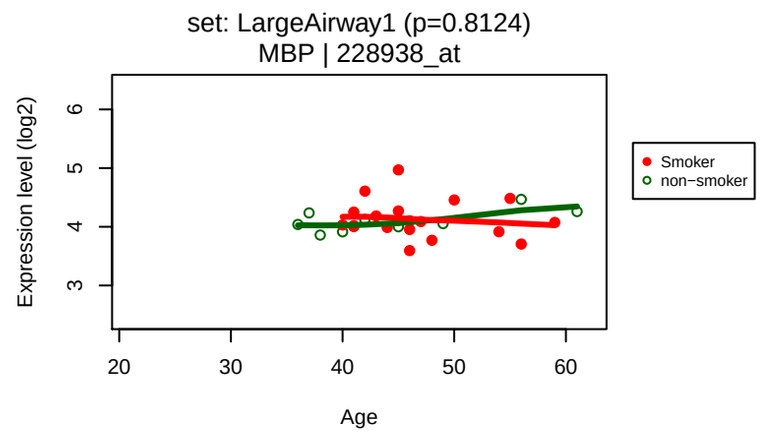

Gene expression of smokers and nonsmokers
gene: MBP | 228938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1532
p=0.0179
p=0.1727
p=0.8124
p=0.021
p=0.1373
p=0.953
p=0.0691
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

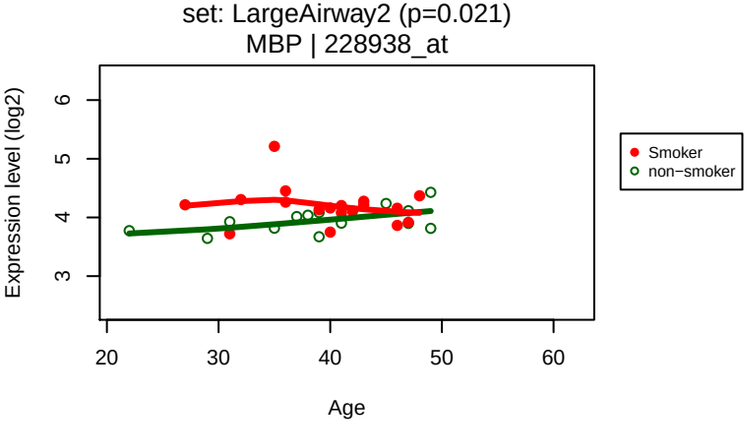

Gene expression ratio (smokers/nonsmokers)
gene: MBP | 228938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1532
p=0.0179
p=0.1727
p=0.8124
p=0.021
p=0.1373
p=0.953
p=0.0691
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

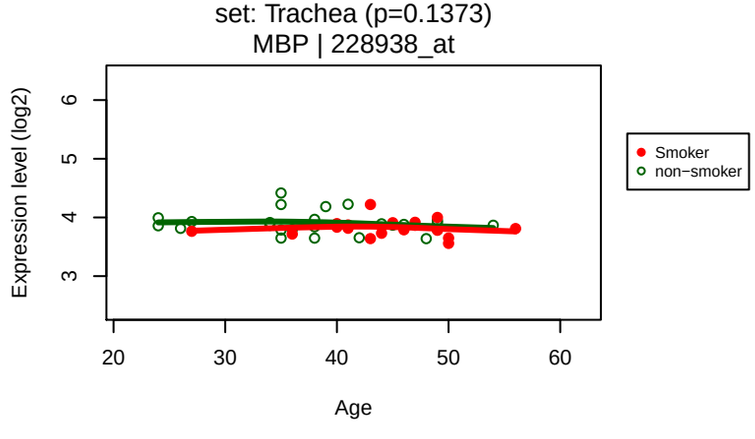

Gene expression ratio in Males
gene: MBP | 228938_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0605
p=0.088
p=0.3832
p=0.0946
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MBP | 228938_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2362
p=0.1947
p=0.0989
p=0.4073
SmallAirway2
LargeAirway2
Trachea
Macrophage2

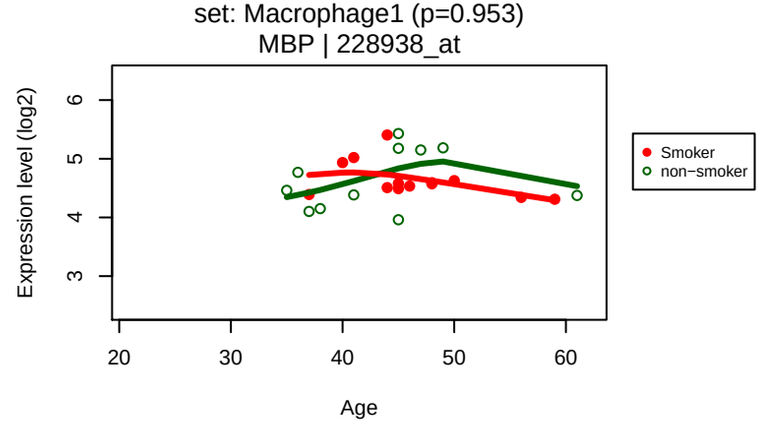

set: SmallAirway1 (p=0.1532)
MBP | 228938_at
Expression level (log2)
Age
Smoker
non−smoker

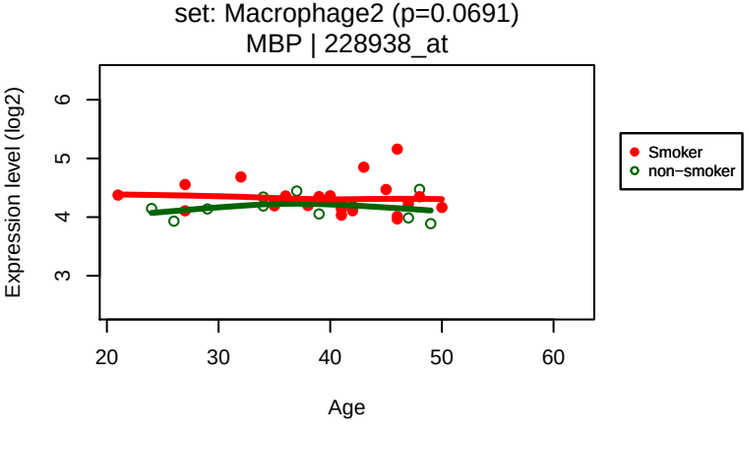

set: SmallAirway2 (p=0.0179)
MBP | 228938_at
Expression level (log2)
Age
Smoker
non−smoker

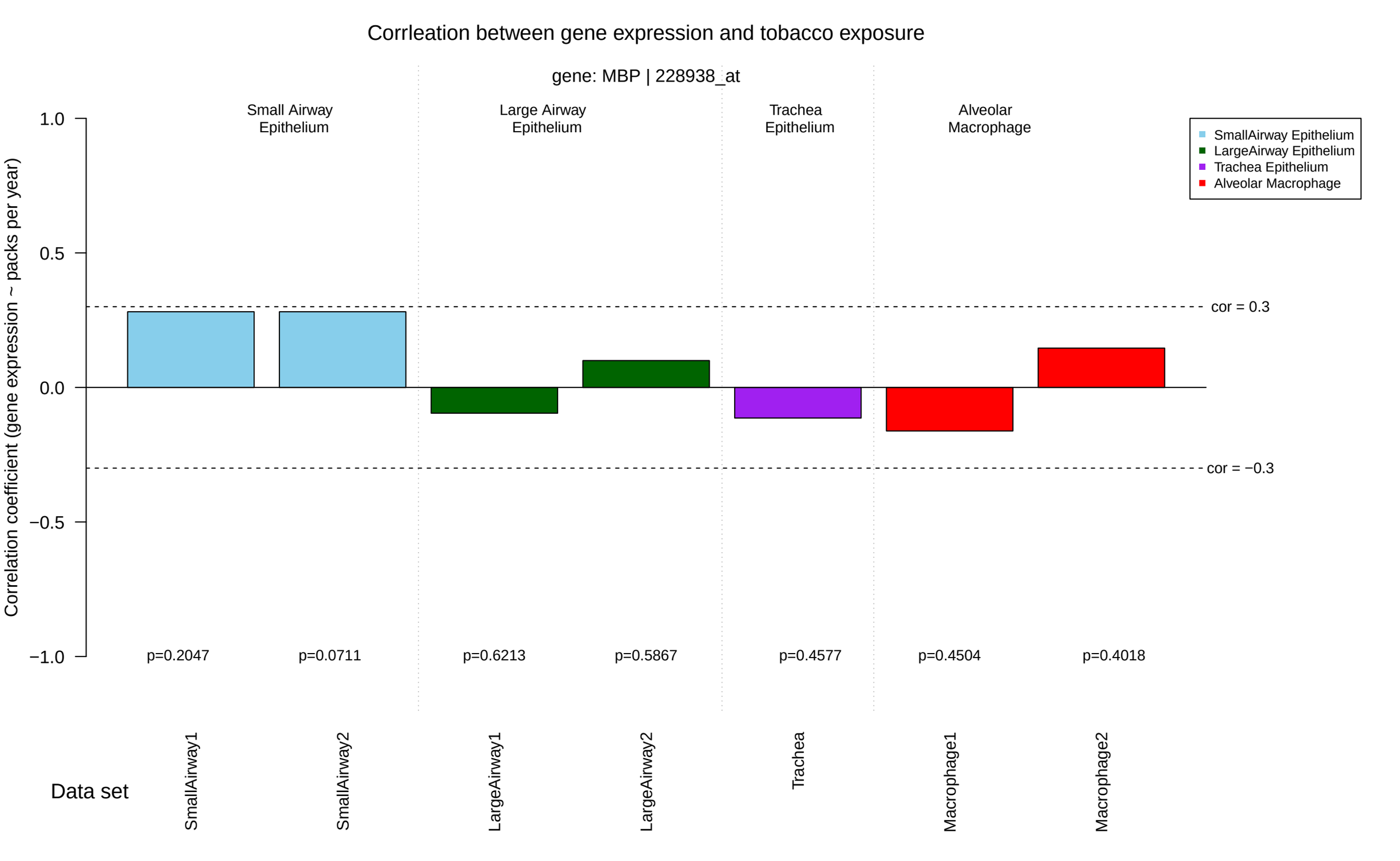

set: LargeAirway1 (p=0.8124)
MBP | 228938_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.021)
MBP | 228938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1373)
MBP | 228938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.953)
MBP | 228938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0691)
MBP | 228938_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MBP | 228938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2047
p=0.0711
p=0.6213
p=0.5867
p=0.4577
p=0.4504
p=0.4018
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2